A Fine-Tuned Language Model for Patient-Aware Drug Interaction Assessment

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Abstract
Drug-drug interactions (DDIs) is an important cause of patient harm, which can be mostly prevented. However, current clinical decision support systems still only consider static database lookups without any reasoning based on patient-specific circumstances. We propose the development and evaluation of fine-tuning a state-of-the-art open-source language model for predicting polypharmacy DDIs. With a basis of 830 known drug-to-drug interactions collected from the DDInter database, we generated a reasoning dataset of synthetic patients and fine-tuned Qwen 3-14B, a modern open-source language model, with low-rank adaptation (LoRA) method. In particular, the proposed method will run locally and comply to patient data protection legislation such as HIPAA in US and KVKK in Turkey for compliance purposes. The fine-tuned model with the non-fine-tuned models are benchmarked using 15 clinical scenarios. The fine-tuned approach reached a success rate of 86.7% (13 out of 15 scenarios) while the baseline achieved 60.0% (9 out of 15 scenarios). Overall, the project attempts to solve data scarcity and patient data privacy issues that can improve the efforts and propose a proof-of-concept method in developing a clinical assist system for DDIs.

1. Introduction

1.1 Problem Definition and Research Challenge

Polypharmacy, which means the use of multiple medications by the same patient, is one of the most overlooked problems in modern clinical practice. As the population gets older and chronic diseases become more common, many patients now use complex medication plans that include five to fifteen drugs at the same time. In addition, the uncontrolled use of over-the-counter vitamins and supplements increase the risk of drug interactions that may have serious clinical consequences.

The clinical risk is important. Drug-drug interactions are estimated to be responsible for 3–5% of in-hospital medication errors and are associated with around 1.1% of hospital admissions in general, with higher rates reported among older adults. In Turkey, one study reported a high prevalence of potential drug interactions, including 81.9% of ICU patients and 34.91% of prescriptions. Datasets such as DDInter and the FDA's DailyMed, or DrugBank contain hundreds of thousands of known interaction pairs with severity levels and short mechanistic explanations, which is a problem already addressed. The real problem to solve is combining these interactions in a patient-specific way. In other words, when a patient has a certain age, weight, health condition, and a list of five to ten medications, what is the overall risk picture, and what clinical recommendations should follow from it?

This difference between pairwise database lookup and patient-level reasoning is the main gap that this project aims to address. A database can easily show that ibuprofen interacts with metformin, but it cannot fully answer in a way that explains this interaction may become more serious in a 68-year-old diabetic patient with an eGFR of 55 mL/min who is also using an ACE inhibitor. It also cannot combine all simultaneous interactions into a clear, prioritized clinical explanation with practical monitoring recommendations. This reasoning gap is the central problem targeted in this work.

1.2 Significance

1.2.1 Patient Safety

Adverse drug reactions are an important cause of preventable patient harm, and drug-drug interactions are one of the factors that contribute to this problem. In the United Kingdom, a prospective study

found that adverse drug reactions were responsible for 6.5% of hospital admissions. More recent evidence also suggests that, among older adults, around 15.1% of ADR-related hospital admissions are caused by clinically important DDIs. Older patients are especially at risk because polypharmacy is common in this group. At the same time, age-related changes in kidney function, liver metabolism, and body composition can change how drugs are absorbed, distributed, metabolized, and removed from the body. These patient-specific changes are not always captured by static severity classifications, which makes simple database-based interaction warnings less reliable in complex clinical situations.

1.2.2 Alert Fatigue in Clinical Settings

Even when clinicians have access to interaction databases, it is still difficult to evaluate every possible drug pair in a complex medication list. For example, a ten-drug regimen creates 45 unique drug pairs, and checking all of them during a normal clinical encounter can be error-prone. Decision support tools often show flat lists of flagged interactions without clear prioritization or explanation. Instead of helping clinicians, this can increase alert fatigue. Published studies report that clinicians override up to 90% of automated drug interaction alerts, mostly because many alerts do not include enough patient-specific context or practical guidance.

For this reason, a system that can produce a synthesized, prioritized, and patient-specific risk narrative would directly address this problem. It would not only show possible interactions, but also explain which ones matter most and what actions should be considered.

1.2.3 The Case for Local Language Model Deployment

The drug interaction domain is suitable for language model approaches because the knowledge in this area is mostly linguistic and relational. Interaction mechanisms are usually explained in natural language in pharmacology literature, drug labels, and clinical guidelines. The expected output is also linguistic: a reasoned and prioritized explanation that a clinician can read and use in practice. However, patient medication lists, laboratory values, and comorbidity profiles include protected health information under regulations such as HIPAA in the United States and KVKK in Turkey. Sending this data to cloud-based frontier models is legally and ethically problematic in most clinical settings without clear data processing agreements. It is also practically difficult in many hospital environments. This limitation is the main reason behind the architectural choice of this project: a fine-tuned, open-source model that runs completely on local infrastructure. In this way, no patient data leaves the institution’s own systems, and the model can provide patient-specific reasoning while keeping sensitive clinical data inside the hospital environment.

2. Related Work and Background

2.1 Structural Limitations of Existing Database Systems

Existing clinical decision support tools are mostly database lookup systems. They are good at identifying known pairwise drug interactions, but they are not designed for holistic reasoning across multiple drugs or for adapting their output to patient-specific parameters.

For example, the interaction between Drug A and Drug B is usually evaluated separately from the interaction between Drug B and Drug C. This can be a problem when both interactions affect the same physiological pathway and may strengthen each other. Also, severity labels in these systems are usually static. A “moderate” interaction may receive the same label for a healthy 30-year-old patient and for a 75-year-old patient with impaired renal function, even though the clinical meaning can be very different in each case.

Because of this, these tools can identify possible risks, but they cannot always explain how the risks combine in a real patient situation.

2.2 Limitations of General-Purpose Large Language Models

Frontier language models have broad pharmacological knowledge because they are trained on large amounts of medical and scientific text. For this reason, they can often produce drug interaction explanations that sound clinically reasonable. However, their architecture and deployment requirements create important limitations for clinical decision support.

This can be explained in two reasons, firstly, these models can still hallucinate specific claims about drugs. When they process long and complex medication lists, they may confuse the relationships between different drugs. For example, a model may correctly recognize a drug name but connect it to a mechanism that actually belongs to another drug in the same prompt. Complex pharmacological names can also be difficult for the model to process, which may cause incorrect severity levels or non-existent metabolic pathways to be generated. Second, these models are not designed to provide patient-specific risk assessment in a structured and reproducible way. Their answers can change depending on small differences in prompt wording, which makes clinical validation difficult across different patient cases. In addition, frontier models usually require cloud-based deployment because of their large computational needs. Sending protected health information, such as medication lists, laboratory values, and comorbidity profiles, to external servers creates serious privacy concerns and is not suitable for many clinical environments.

For these reasons, this project requires a specialized model that can provide strong reasoning while remaining small enough to run on local, air-gapped clinical infrastructure. This approach would allow patient-specific analysis without sending sensitive data outside the institution.

2.3 Absence of a Task-Specific Dataset

Training a specialized model for polypharmacy reasoning requires a dataset that connects patient-specific medication scenarios with verified clinical reasoning outputs. These outputs should not only mention possible drug interactions, but also explain the mechanisms behind them in a medically grounded way. However, publicly available datasets do not have the exact structure needed for this task. They usually do not combine a medication list, patient profile, and an integrated interaction reasoning narrative in the same example. This lack of suitable data has been one of the main bottlenecks preventing fine-tuned specialized models from being used for this problem.

The work of Tatonetti et al. (2012) showed the potential of data-driven methods for predicting drug-drug interactions. However, their work focused mainly on detecting signals from adverse event reports, rather than generating patient-level reasoning for complex medication cases.

2.4 Prior Work on Fine-Tuning for Medical Tasks

Low-rank adaptation techniques, especially LoRA (Hu et al., 2022), have shown that large language models can be adapted to specialized tasks with much lower computational requirements. This makes it possible to fine-tune domain-specific models on consumer hardware and supports the idea of local clinical deployment.

Several studies have used instruction fine-tuning for medical question-answering tasks. However, the specific problem of contextualized polypharmacy reasoning, using grounded and verifiable training data, remains mostly unaddressed in the published literature. This creates an important opportunity for developing a smaller specialized model that can reason over patient-specific medication scenarios while still being feasible to run on local infrastructure.

3. Methodology

3.1 Dataset Source: DDInter

The factual foundation for the training dataset is the DDInter database, an open-access drug-drug interaction repository containing over 236,000 DDI records. DDInter was chosen over alternatives such as DrugBank primarily due to access constraints, whereas DDInter is freely accessible and provides structured DDI records with mechanism descriptions and severity classifications.

The DDInter database was crawled to extract drug-drug interaction records. After deduplication and filtering for records with sufficient mechanistic description, 830 unique drug-drug interaction pairs were retained as the factual seed corpus. Each record includes the two interacting drugs, a severity classification, and a mechanistic description of the interaction. This corpus was not used as training data directly, DDInter served as a verified factual layer for the data generation step.

3.2 Training Dataset Construction via Grounded Generation

A main methodological challenge in this domain is that no existing dataset connects patient profiles with integrated polypharmacy reasoning outputs. To address this gap, a synthetic dataset construction pipeline was developed using a grounded generation approach. In this pipeline, the factual content is taken only from DDInter-verified records, while a frontier language model is used only to create and format the clinical narrative.

For each DDI pair in the seed corpus, a synthetic patient profile was created to represent a realistic clinical scenario. To make the dataset more robust, the generation pipeline was instructed to vary demographic features such as age and sex. It was also instructed to include specific informations that can increase risk, such as different stages of chronic kidney disease measured by eGFR, or hepatic impairment. Then, a frontier model was guided by a strict system prompt. The model had to generate a structured clinical reasoning narrative based only on the given DDInter entry and patient profile. The prompt required a standardized JSON output with four keys, priority interactions, mechanistic explanation, patient-specific risk assessment, and monitoring recommendations, such as:

[1. Priority Interactions]

Moderate interaction between Amphotericin B (cholesteryl sulfate) and Rabeprazole.

[2. Mechanistic Explanation]

Chronic use of proton pump inhibitors (PPIs) like rabeprazole can induce hypomagnesemia. The risk is increased when PPIs are used concomitantly with other agents that can cause magnesium loss, such as amphotericin B. Both drugs can contribute to magnesium depletion.

[3. Patient-Specific Risk Assessment]

This patient is at increased risk for developing hypomagnesemia, which can lead to serious adverse events including seizures, arrhythmias (e.g., atrial fibrillation, supraventricular tachycardia), and abnormal QT interval. The hypomagnesemia may not resolve with magnesium supplementation alone and may require discontinuation of the PPI.

[4. Recommendations]

Monitor serum magnesium levels prior to initiating therapy and periodically thereafter, especially given the prolonged use of amphotericin B and rabeprazole. Advise the patient to seek immediate medical attention if they develop potential signs and symptoms of hypomagnesemia such as palpitations, arrhythmia, muscle spasm, tremor, or convulsions.

This grounded generation approach helps limit hallucination mostly to narrative framing rather than factual content. By anchoring the generation process to verified pharmacological mechanisms, the resulting dataset covers different severity levels and provide the linguistic richness and structural consistency needed for effective instruction fine-tuning.

3.3 Model Selection: Qwen 3-14B

Qwen 3-14B was selected as the base model for fine-tuning after considering both its technical capability and the deployment needs of this project. The Qwen 3 series, developed by Alibaba's Qwen team, is a recent open-source language model family trained on a large multilingual corpus, including scientific and medical texts. Qwen 3 models also support hybrid reasoning, which allows the model to use longer reasoning steps for more complex tasks. This is useful for a clinical application where the model needs to combine several pharmacological relationships before reaching a patient-specific conclusion.

The 14-billion parameter size was chosen as a balance between reasoning ability and local deployment feasibility. Smaller models, especially those below around 7 billion parameters, may struggle with the multi-step reasoning needed to connect patient values with drug interaction risks. On the other hand, models above 30 billion parameters usually require multi-GPU infrastructure, which does not fit the privacy-focused and locally deployable design of this system. Therefore, the 14B scale provides a practical middle ground. It has enough reasoning capacity for structured clinical outputs, can be deployed on a single modern GPU with quantization, and can be used on-premise without sending patient data outside the institution.

3.4 Fine-Tuning with Low-Rank Adaptation (LoRA)

Fully fine-tuning a 14-billion parameter model normally requires updating billions of weight values at the same time. This creates high GPU memory and compute requirements. Low-Rank Adaptation, or LoRA, provides a more efficient alternative by freezing the original pre-trained model weights and adding small trainable adapter matrices into the model architecture. Instead of changing the whole model, LoRA trains only these additional parameters, which makes the process much lighter and more suitable for local fine-tuning.

The training was implemented using the Unsloth library in order to optimize VRAM usage. The selected LoRA configuration was python r = 32, target_modules = ["q_proj", "k_proj", "v_proj", "o_proj", "gate_proj", "up_proj", "down_proj"].

With this configuration, the number of trainable parameters was 128,450,560 out of 14,896,757,760 total parameters, meaning that only 0.86% of the model parameters were updated during training. This shows the efficiency of the LoRA-based approach, since the model could be adapted to the task without modifying the vast majority of its original weights.

The rank value was set to 32 to give the adapter layers enough capacity to learn the structured response format and patient-specific reasoning patterns required by the task. Since the objective was not only to teach a simple output style, but also to guide the model in reasoning over drug pairs, severity levels, mechanisms, patient conditions, and clinical recommendations, a relatively higher LoRA rank was preferred. The target modules were selected from both attention and feed-forward layers, allowing the adaptation process to influence both information selection and internal representation transformation during generation.

In addition to the LoRA configuration, the supervised fine-tuning process was controlled using a memory-efficient training setup. The per-device training batch size was set to 2, and gradient accumulation was performed over 4 steps. This resulted in an effective batch size of 8 while keeping the number of examples processed simultaneously on the GPU low. This was important for making the training feasible under local hardware constraints while still producing more stable parameter updates than a very small batch size would allow.

Prior to billion-parameter models, the usual approach while fine-tuning was to train the non-frozen layers of the model for a few epochs to learn new features. However, since models such as Qwen have

billions of parameters, and the datasets are relatively large, the standard method is often to train using a limited number of optimizer steps, sometimes below one full epoch, rather than repeatedly passing through the entire dataset. Since our dataset is relatively small, the chosen approach was to fine-tune the model for one full epoch over the dataset to cover all interaction pairs, without skipping. Since the dataset contained 830 examples and the effective batch size was 8, one epoch corresponded to approximately 104 optimizer updates. Training for one epoch ensured that the model was exposed to every training example at least once, which was important because the dataset contained source-grounded drug-drug interaction information. However, the purpose of this fine-tuning was not to make the model memorize all possible interaction facts. Instead, the goal was to adapt the model to the expected clinical reasoning structure and teach it how to organize patient-specific polypharmacy risk assessments in a consistent, source-aware format.

A learning rate of 2×10^{-4} was used during training. This value was selected because LoRA updates only a small number of adapter parameters rather than the full model weights, allowing a higher learning rate than would usually be appropriate for full-model fine-tuning. A linear learning rate scheduler was used with 5 warmup steps, allowing the optimization process to begin gradually before reaching the main learning rate. This helped reduce instability during the early part of training.

The AdamW 8-bit optimizer was used, which is a memory-efficient version of AdamW. A small weight decay value of 0.001 was also used to provide regularization and reduce unnecessary growth in the adapter weights. Overall, this setup made it possible to fine-tune the Qwen 3-14B model locally with minimal VRAM requirements without full-model training, which would be close to impossible for the term project.

3.5 Evaluation Protocol

The fine-tuned model was evaluated against the non-fine-tuned Qwen 3-14B baseline on a set of 15 clinical test cases. Each test case consisted of a patient profile and a medication list requiring drug interaction assessment. These test cases were not used during training, so they provided a separate way to measure whether fine-tuning improved the model's ability to reason over new patient scenarios. A test case was considered a pass if the model correctly identified the clinically significant interaction pairs, provided a mechanistically accurate explanation grounded in the DDInter data, and produced an appropriate patient-contextualized risk assessment. This means that the model was not evaluated only on whether it mentioned a possible interaction, but also on whether it explained the interaction correctly and connected it to the patient's specific clinical parameters. A fail was recorded when the model missed a significant interaction, produced an incorrect mechanism, or provided a risk narrative that was inconsistent with the patient's clinical values. For example, if the model failed to recognize increased risk in a patient with reduced renal function, or if it attributed a drug interaction to the wrong metabolic pathway, the case was marked as a failure. This evaluation protocol was designed to compare the practical reasoning behavior of the fine-tuned model with the original base model. The aim was to test whether fine-tuning improved not only formatting consistency, but also the model's ability to generate accurate, structured, and patient-specific polypharmacy reasoning.

4. Results

4.1 Quantitative Evaluation

The fine-tuned model and the non-fine-tuned baseline were both evaluated across the same 15 clinical test cases. The results are shown in Table 1.

Model	Pass	Fail	Pass Rate
Qwen 3-14B (Non-Fine-Tuned Baseline)	9	6	60.0%
Qwen 3-14B (Fine-Tuned, DDInter LoRA)	13	2	86.7%

Table 1. Pass/fail evaluation results across 15 clinical test cases.

The fine-tuned model achieved a pass rate of 86.7% (13 of 15), compared to 60.0% (9 of 15) for the non-fine-tuned baseline. This represents an improvement of 26.7 percentage points, reducing the number of failed test cases from 6 to 2. The result demonstrates that the LoRA fine-tuning on the DDInter-grounded synthetic dataset produced a meaningful and measurable improvement in the model's ability to perform patient-contextualized drug interaction reasoning.

Some examples from the fine-tuned model, along with the user's requests:

1) **User**: I am treating a 74-year-old male with type 2 diabetes and stable heart failure. He is on carvedilol 6.25 mg BID and furosemide 20 mg daily. We are planning to start semaglutide 0.25 mg weekly. If we also introduce mesoridazine for his psychiatric symptoms, what are the concerns regarding his glycemic control?

Model Response:

[1. Priority Interactions]

Mesoridazine and Semaglutide

[2. Mechanistic Explanation]

Mesoridazine, a phenothiazine antipsychotic, can cause hyperglycemia and diabetes mellitus. This effect is thought to be due to its interference with glucose metabolism and its potential to increase insulin resistance.

When combined with semaglutide, which is a GLP-1 receptor agonist used to treat type 2 diabetes, the risk of hyperglycemia may be increased. Semaglutide works by enhancing insulin secretion and reducing glucagon secretion, which helps lower blood glucose levels.

However, the use of mesoridazine could counteract this effect by increasing glucose production and reducing glucose utilization, potentially leading to elevated blood glucose levels.

[3. Patient-Specific Risk Assessment]

This 74-year-old male patient has type 2 diabetes and is currently on semaglutide 0.25 mg weekly, which is used to manage his condition. The introduction of mesoridazine for psychiatric symptoms raises concerns about his glycemic control.

Mesoridazine's potential to cause hyperglycemia could interfere with the effectiveness of semaglutide, leading to suboptimal glucose control. This is particularly important given his age and existing

diabetes, as uncontrolled hyperglycemia can lead to serious complications such as diabetic ketoacidosis, neuropathy, and retinopathy.

His heart failure status also needs to be considered, as hyperglycemia can exacerbate cardiovascular conditions.

[4. Recommendations]

Close monitoring of blood glucose levels is essential for this patient when mesoridazine is introduced. His semaglutide dosage should be adjusted as necessary to maintain adequate glycemic control.

The healthcare provider should review his current diabetes management plan and consider any adjustments to his antidiabetic medications.

If his blood glucose levels rise significantly, the provider should evaluate the possibility of discontinuing mesoridazine or adjusting the dose of semaglutide.

Patient education on the importance of glucose monitoring and the potential side effects of mesoridazine is also crucial.

2) User: I am conducting a medication review for an 82-year-old male with obstructive sleep apnea and chronic back pain. His current medications include paramethadione, primidone, hydromorphone, methsuximide, ritodrine, and metipranolol eye drops. His recent clinical values are O2 saturation 92%, heart rate 60 bpm, and creatinine 1.1 mg/dL. Can you identify the source-supported drug-interaction risks in this list and prioritize monitoring?

Model Response:

[1. Priority Interactions]

- Paramethadione + Primidone: Moderate. Moderate interaction identified between paramethadione and primidone.

- Hydromorphone + Methsuximide: Moderate. Moderate interaction identified between hydromorphone and methsuximide.

- Ritodrine + Metipranolol eye drops: Moderate. Moderate interaction identified between ritodrine and metipranolol eye drops.

These are the source-supported pairwise risks in this synthetic case; other medication pairs in the list are not being labeled unless listed above.

[2. Mechanistic Explanation]

The main theme of this case is cns/respiratory depression: focus on CNS depression, sedation, respiratory monitoring, falls risk, and source-supported caution.

- Paramethadione + Primidone: Primidone may potentiate the central nervous system (CNS) and respiratory-depressant effects of paramethadione. Both are CNS-acting agents, and their combined use can lead to excessive or prolonged CNS and respiratory depression, especially in elderly or debilitated patients.

- Hydromorphone + Methsuximide: The concomitant use of hydromorphone, an opioid, with methsuximide, a barbiturate, may result in additive or synergistic central nervous system (CNS) and respiratory-depressant effects. This combination increases the risk of profound sedation, coma, and respiratory depression, particularly in elderly or debilitated patients.

- Ritodrine + Metipranolol eye drops: The use of ritodrine with beta-blockers like metipranolol can

lead to additive or synergistic hypotensive effects, potentially causing dizziness, syncope, and orthostatic hypotension. Ritodrine's beta-agonist effects can be counteracted by beta-blockers, increasing the risk of cardiovascular compromise.

[3. Patient-Specific Risk Assessment]

This 82-year-old male has obstructive sleep apnea and chronic back pain, with recent values of O2 saturation 92%, heart rate 60 bpm, and creatinine 1.1 mg/dL. These patient factors make the listed source-supported interactions clinically important to review together. The combined regimen should be treated as a polypharmacy-risk scenario, but the assessment should remain bounded to the listed source-supported pairs rather than inventing additional interactions.

[4. Recommendations]

Prioritize the highest-severity and mechanism-convergent risks first. Review whether any interacting drugs are avoidable, correct modifiable risk factors, and monitor the patient using the source-supported recommendations below:

- Paramethadione + Primidone: Caution is advised when combining paramethadione with primidone. Monitor the patient closely for excessive or prolonged CNS and respiratory depression, especially during the initial dosing period. Consider reducing the dosage of paramethadione or avoiding its use if the patient is elderly or debilitated. Educate the patient to avoid hazardous activities requiring mental alertness and motor coordination.

- Hydromorphone + Methsuximide: Caution is advised if hydromorphone is used with methsuximide. Monitor the patient closely for signs of excessive or prolonged CNS and respiratory depression, such as profound sedation, confusion, and slowed or shallow breathing. Dosage adjustments of hydromorphone may be required, particularly in elderly or debilitated patients. Educate the patient to avoid hazardous activities requiring mental alertness and motor coordination.

- Ritodrine + Metipranolol eye drops: Use caution when ritodrine is used in combination with beta-blockers like metipranolol. Monitor the patient closely for symptoms of orthostatic hypotension, such as dizziness, lightheadedness, syncope, and blurred vision. Advise the patient to avoid rising abruptly from a sitting or recumbent position and to notify their physician if they experience these symptoms.

4.2 Observations

Beyond the aggregate pass rate, qualitative comparison of model outputs revealed consistent behavioral differences between the baseline and the fine-tuned model. The baseline model tended to produce generic interaction warnings without integrating patient-specific parameters such as renal function or age-related pharmacokinetic changes. In several cases, the baseline correctly identified that an interaction existed but failed to contextualize its severity relative to the patient profile, which was scored as a fail under the evaluation criteria.

The fine-tuned model consistently applied patient context to modulate interaction severity. In test cases involving renally impaired patients, the fine-tuned model appropriately escalated the risk assessment for interactions involving renally cleared drugs, producing outputs structurally similar to the training examples. The model also demonstrated improved output formatting, consistently producing the four-part structured output (priority interactions, mechanistic explanation, patient-specific risk assessment, and monitoring recommendations) that was enforced during dataset construction.

5. Discussion and Conclusion

5.1 Discussion

The results support the core hypothesis of this work: that fine-tuning on a relatively small grounded synthetic dataset can substantially improve a language model's ability to reason about drug interactions in a patient-specific manner. The 26.7 percentage point improvement in pass rate with only 830 source interactions and a compact LoRA adapter suggests that the task-specific reasoning pattern, rather than factual memorization, is the primary benefit of the fine-tuning process.

The grounded generation methodology appears to have been effective in constraining hallucination. No test case failures were attributed to factually incorrect interaction mechanisms in the fine-tuned model's outputs; the two remaining failures were related to incomplete coverage of interaction pairs in complex multi-drug scenarios. This is consistent with the design intent: factual grounding in DDInter prevents the model from inventing mechanisms, while the synthetic patient profiles teach it to apply those mechanisms contextually.

Several limitations should be acknowledged. The evaluation set of 15 test cases is small. Furthermore, the training dataset, while grounded in verified DDInter records, relies on synthetically generated patient profiles and narratives that is reviewed manually, but not been reviewed by clinical pharmacologists. The system should therefore be considered a proof-of-concept demonstration that could be expanded into a clinically usable local language model approach rather than a clinically validated tool. Expanding the training dataset to include reasoning steps may include benefits to the model results through complex scenarios as well.

5.2 Conclusion

This work presents a fine-tuned language model for patient-contextualized drug interaction risk reasoning. It addresses the main gap between pairwise drug interaction databases and the patient-level synthesis required in clinical practice. The approach combines DDInter-grounded synthetic data construction with LoRA-based fine-tuning of Qwen 3-14B, resulting in a locally deployable system designed for clinical environments where patient privacy is a major constraint.

The fine-tuned model improved from a 60.0% pass rate to an 86.7% pass rate on a 15-case clinical evaluation, providing initial evidence that the methodology is effective. The two main contributions of this work are, first, a grounded dataset construction methodology for a clinical reasoning task where suitable training data is scarce, with a clear separation between factual sourcing and narrative generation to reduce hallucination and a demonstration of patient-context sensitivity in a fine-tuned language model for local deployment, where the same drug pairs can produce different and appropriate risk narratives depending on patient parameters.

Future work should focus on expanding the evaluation dataset, including clinical expert review of the training and evaluation examples, potentially extending the coverage to drug-food interactions and supplement interactions, and conducting prospective evaluation in a clinical environment. The longer-term goal is to build a system that can be integrated into electronic health records as a locally deployable, privacy-preserving polypharmacy risk advisor.

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